

Chapter 13

Energy with dielectrics

13.1 Work in the presence of dielectrics

We want the work needed to bring free charges from infinity in the presence of dielectric material. Changing the free charge density by $\delta\rho$ costs

$$\delta W = \int \delta\rho \phi dV \quad , \quad 4\pi \delta\rho = \vec{\nabla} \cdot \delta\vec{D} \quad (13.1)$$

so that

$$\begin{aligned} \delta W &= \frac{1}{4\pi} \int (\vec{\nabla} \cdot \delta\vec{D}) \phi dV \\ &= -\frac{1}{4\pi} \int \delta\vec{D} \cdot \vec{\nabla} \phi dV = \frac{1}{4\pi} \int \vec{E} \cdot \delta\vec{D} dV \end{aligned} \quad (13.2)$$

integrating by parts. The total work builds up the fields from zero

$$W = \int_0^D \delta W \quad (13.3)$$

Assume a linear material, so that

$$\vec{E} \cdot \delta\vec{D} = \frac{1}{2} \delta (\vec{E} \cdot \vec{D}) \quad (13.4)$$

which gives

$$W = \frac{1}{8\pi} \int \vec{E} \cdot \vec{D} dV \quad (13.5)$$

i.e. an energy density $\vec{E} \cdot \vec{D}/8\pi$ instead of $E^2/8\pi$.

13.2 Energy change from adding material

Consider next the energy change from adding dielectric material with no changes to the free charges

$$\Delta W = \frac{1}{8\pi} \int \left(\vec{E} \cdot \vec{D} - \vec{E}_0 \cdot \vec{D}_0 \right) dV \quad (13.6)$$

where $\vec{E}_0$, $\vec{D}_0$ are the fields before the material is added. Rewrite the integrand as

$$\vec{E} \cdot \vec{D} - \vec{E}_0 \cdot \vec{D}_0 = \vec{E} \cdot \vec{D}_0 - \vec{D} \cdot \vec{E}_0 + \left(\vec{E} + \vec{E}_0 \right) \cdot \left(\vec{D} - \vec{D}_0 \right) \quad (13.7)$$

The last term integrates to zero: $\vec{E} + \vec{E}_0 \propto \vec{\nabla} \phi$ while $\vec{\nabla} \cdot (\vec{D} - \vec{D}_0) = 0$, since the free charges are unchanged. Hence

$$\begin{aligned} \Delta W &= \frac{1}{8\pi} \int \left(\vec{E} \cdot \vec{D}_0 - \vec{D} \cdot \vec{E}_0 \right) dV \\ &= -\frac{1}{8\pi} \int (\epsilon - \epsilon_0) \vec{E} \cdot \vec{E}_0 dV \end{aligned} \quad (13.8)$$

using $\vec{D} = \epsilon \vec{E}$ and $\vec{D}_0 = \epsilon_0 \vec{E}_0$. The corresponding pressure on the material is

$$\text{Pressure} = \frac{1}{8\pi} (\epsilon - \epsilon_0) \vec{E} \cdot \vec{E}_0 \quad (13.9)$$

Note that if $\epsilon_0 = 1$, we get

$$\Delta W = -\frac{1}{2} \int \vec{P} \cdot \vec{E}_0 dV \quad (13.10)$$

like a dipole energy.

13.3 Example: capacitor at fixed charge

Consider a parallel-plate capacitor with plate charges $-Q$ and $+Q$ and separation d , with a dielectric slab ϵ filling the left portion of the gap and vacuum $\epsilon_0 = 1$ on the right, as depicted in the Figure; a is the total plate area and a' is the area of the left side, covered by the dielectric. The surface charge densities are $\mp\sigma'$ over the dielectric and $\mp\sigma$ over the vacuum region. In the

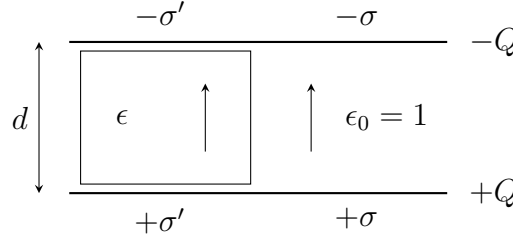


Figure 13.1: A parallel-plate capacitor with plate charges $\mp Q$ and separation d ; a dielectric slab ϵ fills the left portion of the gap, of plate area a' out of the total area a , with vacuum $\epsilon_0 = 1$ on the right. The surface charge densities are $\mp\sigma'$ over the dielectric and $\mp\sigma$ over the vacuum region.

vacuum region and in the dielectric respectively,

$$D = 4\pi\sigma \Rightarrow E = 4\pi\sigma \quad , \quad D' = 4\pi\sigma' \Rightarrow E' = \frac{4\pi\sigma'}{\epsilon} \quad (13.11)$$

The boundary condition equates the two fields

$$E = E' \Rightarrow \sigma = \frac{\sigma'}{\epsilon} \quad (13.12)$$

The total charge on a plate is then

$$Q = \sigma'a' + \sigma(a - a') = \sigma((\epsilon - 1)a' + a) \Rightarrow \sigma = \frac{Q}{(\epsilon - 1)a' + a} \quad (13.13)$$

With $E_0 = 4\pi Q/a$ the field before the dielectric is added, (13.8) gives

$$\begin{aligned} \Delta W &= -\frac{1}{8\pi} \left[a' d E' \frac{4\pi Q}{a} (\epsilon - 1) + (a - a') d (\epsilon - 1) E \frac{4\pi Q}{a} \right] \\ &= -\frac{2\pi Q^2 d (\epsilon - 1)}{(\epsilon - 1)a' + a} \end{aligned} \quad (13.14)$$

which for $a' = a$ becomes

$$\Delta W = -2\pi Q^2 d \left(\frac{1}{a} - \frac{1}{\epsilon a} \right) = -\frac{Q^2}{2C} + \frac{Q^2}{2C'} < 0 \quad (13.15)$$

in terms of the capacitance $C = a/(4\pi d)$ of the empty capacitor and $C' = \epsilon C$: the energy is lowered, *i.e.* the dielectric is attracted into the capacitor.

13.4 Example: capacitor at fixed potential

Now instead keep the potential difference constant. Then we cannot use ΔW from before (13.8), since the free charges change. Instead, use directly

$$W = \frac{1}{8\pi} \int \vec{E} \cdot \vec{D} dV \quad (13.16)$$

with $\vec{D} = \epsilon \vec{E}$ and $\vec{E}$ unchanged. Take an $\ell \times \ell$ capacitor with $\Delta\phi = \text{constant}$, plate separation d , and the dielectric inserted a distance x into the gap, as in the Figure. Then

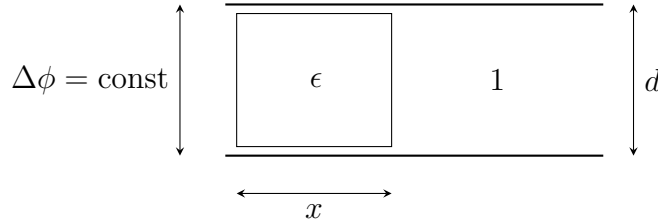


Figure 13.2: An $\ell \times \ell$ parallel-plate capacitor held at fixed potential difference $\Delta\phi$, with plate separation d and a dielectric slab ϵ inserted a distance x into the gap; the remaining region is vacuum.

$$W = \frac{1}{8\pi} E^2 [(\ell - x) \ell d + \epsilon \ell x d] \quad (13.17)$$

and

$$\begin{aligned} \frac{dW}{dx} &= \frac{E^2}{8\pi} [-\ell d + \epsilon \ell d] \\ &= \frac{(\Delta\phi/d)^2}{8\pi} (\epsilon - 1) \ell d = \text{Force} \end{aligned} \quad (13.18)$$

the force pulling the slab into the capacitor.